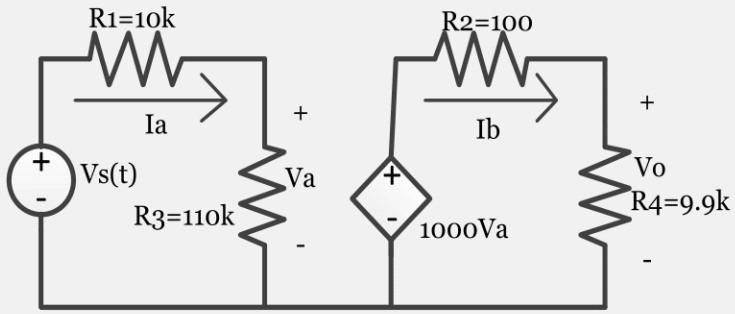
Bangladesh Telecommunication Company Limited

September - 2015

Exam Venue- BUET

Question 1: If Vs(t) = 230sin314t mv, Find Vo.



Reference: Chapter 4 "Fundamentals of Electric Circuits" – Alexander Sadiku

Solution 1:

Vs(t) = 230sin314t mv = $.230\angle(-90)$

In left loop,

Vs – IaR1 – IaR3 = 0

=>Ia = Va/(R1+R3)

=>Ia = $.230\angle(-90)/(10+110)k$

=>Ia = $1.917\times10^{-6}\angle(-90)$

So, Va = IaR3

=>Va = $1.917\times10^{-6}\angle(-90)\times110K$

=>Va = $.211\angle(-90)$

Say, Vx = 1000Va

So, Vx = $210.833\angle(-90)$

Now in right loop,

Vx – IbR2 – IbR4 = 0

=>Ib = Vx/(R2+R4)

=>Ib = $210.833\angle(-90)/(100+9900)$

=>Ib = $.021\angle(-90)$

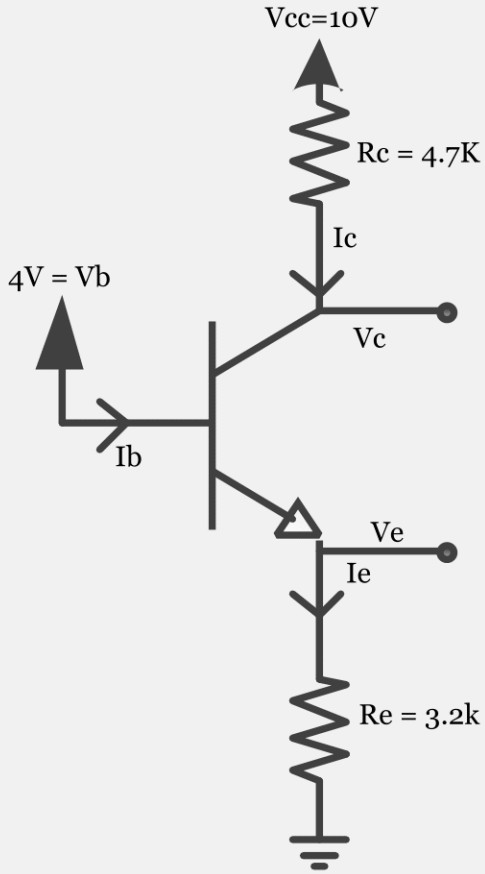
So, Vo = IbR4

=>Vo = $.021\angle(-90)\times9900$

=> Vo = $208.72\angle(-90)$ **(Answer)**

.

Question 2:Find all branch current and node voltages for β = 100



Reference: Chapter 4,"Electronic Devices & Circuit Theory", Boylestad

Solution 2:

Here, Vb = 4V(Answer)

At base emitter loop

Vb – Vbe – IeRe = 0

=> 4 – 0.7 – Ie×(3.2k) = 0

=>Ie = 3.7/3.2k = 1.03125mA

=>Ie = 1.03125mA(Answer)

Ic = α Ie

=>Ic = β/(1+β) × Ie

=>Ic = (100 × 1.0312)/101 = 1.021 mA (Answer)

Ic + Ib = Ie

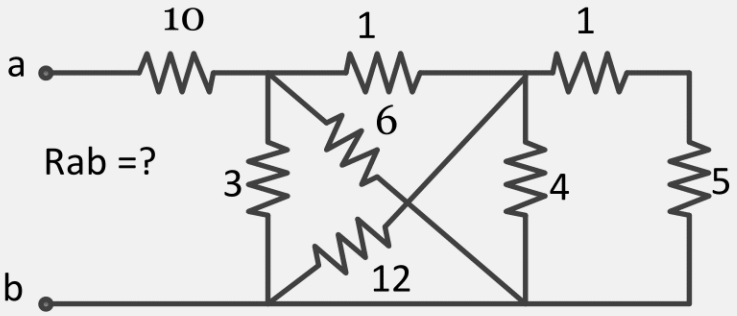
=>Ib = Ie – Ic = 1.031 – 1.02 = .01 mA (Answer)

Now,

Vcc – IcRc – Vc = 0

	=>Vc = 10 - (1.021×4.7) = 5.206V(Answer) Again Ve – IeRe = 0 Ve = 1.03125 × 3.2 = 3.3V(Answer)

Question 3:Find the equivalent resistance of the following circuit.



Reference: Chapter 2, "Fundamentals of Electric Circuits" – Alexander Sadiku

Solution: 3

$$Rab = [[[[\{(5+1) \,||\, 4\} \,||\, 12]+1] \,||\, 6] \,||\, 3] + 10$$

$$= 11.2\ \Omega$$

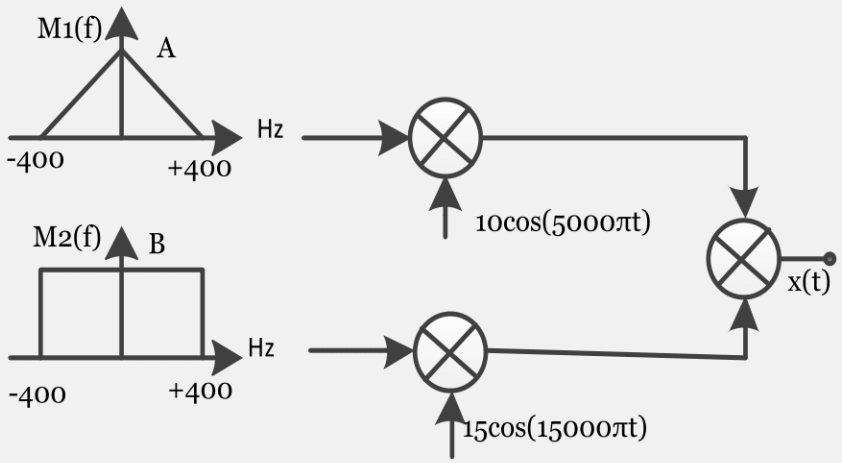
Question 4: If the current of a silicon diode with Vt = 26mV doubles, find the increase in forward voltage drop, $Ip = Io\ e^{Vd/nVt}$

Reference: Chapter 1,"Electronic Devices & Circuit Theory", Boylestad

Solution: 4 Here,	Now doing ln in both the sides $\ln(2) = Vd1 - Vd2 / nVt$

$I_1 = I_o e^{Vd1/nVt}$ ----------[1] $I_2 = 2\,I_1 = I_o e^{Vd2/nVt}$ ---------- [2] Doing equation [2] / equation [1] $2 = e^{(Vd1 - Vd2)/nVt}$	$\Rightarrow Vd1 - Vd2 = \ln(2) \times n \times Vt$ $\Rightarrow Vd1 - Vd2 = .69 \times 2 \times 26mV = 36.04$ mV **(Answer)** [n =2 for silicon diode]

Question 5: Draw the spectrum of x(t)



Reference: Chapter 3,"Modern Digital & Analog Communication Systems", B.P.Lathi

Solution: 5

$$g(t)\cos\,\omega_0 t = \frac{1}{2}\left[g(t)e^{j\omega_0 t} + g(t)e^{-j\omega_0 t}\right]$$

If we do fourier transform, then it becomes

$$g(t)\cos\,\omega_0 t \Longleftrightarrow \frac{1}{2}\left[G(\omega - \omega_0) + G(\omega + \omega_0)\right]$$

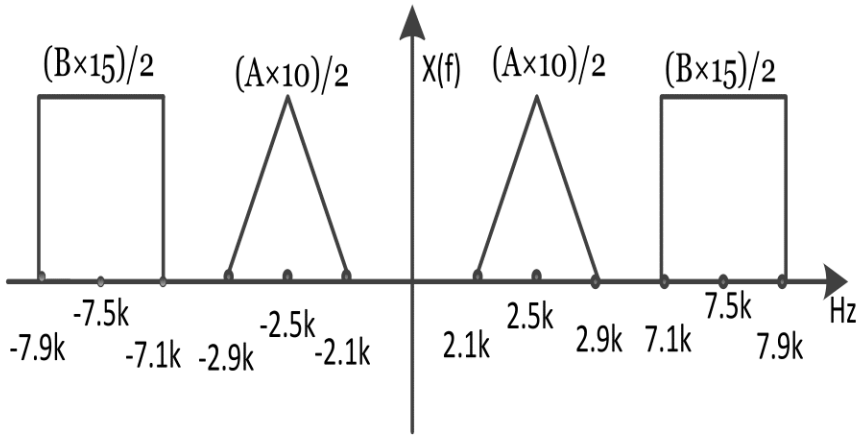
After modulation the center frequency of m1(t) is 2.5kHz, -2.5khz and center frequency of m2(t) is 7.5kHz , -7.5kHz.

The corner requency of m1(t) = (2.5+.4) & (2.5-.4) and m2(t) = (7.5+.4)& (7.5-.4)

Since, the amplitude of M1(f) is A and amplitude of carrier signal no. 1 is 10, so X(f) will have ampitude will be (A×10)/2 for M1(f) and carrier carrier signal no. 1

Similarly for M2(f) and carrier carrier signal no. 2 the amplitude of X(f) will be (B×15)/2

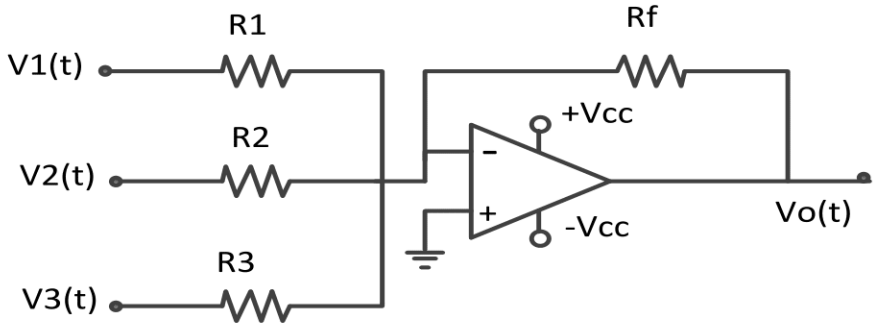
So, the spectrum looks like following



Question 6: Draw circuit diagram of three input summing amplifier.

Reference: Chapter 3,"OP-AMP & Linear IC", Coughlin

Solution: 6



Question 7: Elaborate: PSTN, ADSL, IGN, GSM, TDM

Reference: Different Sources

Solution: 7

PSTN : Public Switched Telephone Network

ADSL : Asymmetric Digital Subscriber line

IGP: Internet Gateway Protocol

GSM : Global System for Mobile communication

TDM: Time Division Multiplexing

Question 8:3signals having frequency of 1200Hz, 600Hz and 600Hz are sampled at Nyquist rate and then multiplexed in a channel 12bit word, find bit rate.

Reference: Chapter 6,"Data Communication & Networking", Behrouz A.Forouzan

Solution: 8

Here, quantization level is not specified, so, $L = 2^n = 2$ assumed

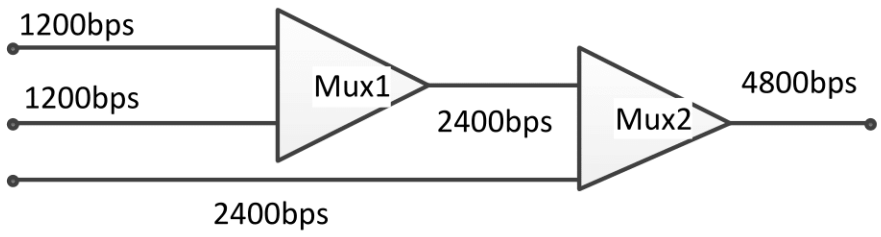
So, n=1

Bit rate of m1(t) = $(2 \times 1200 \times 1) = 2400$bps [n =1]

Bit rate of m1(t) = $(2 \times 600 \times 1) = 1200$bps

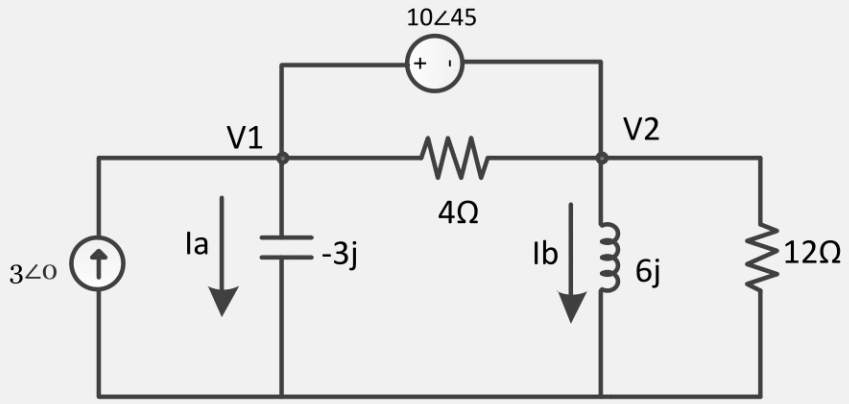
Bit rate of m1(t) = $(2 \times 600 \times 1) = 1200$bps

When multiplexed, the figure looks like this



So, Output data rate = 4800 bps **(Answer)**

Question 9: Find the V1 and V2



Reference: Chapter 10, "Fundamentals of Electric Circuits" – Alexander Sadiku

Solution: 9

$V1 - V2 = 10\angle 45$

In supernode if we apply KCL, then

$3\angle 0 = -V1/3j + V2/6j + V2/12$

$=> 3\angle 0 = .33j \times V1 - .167j \times V2 + .08 \times V2$

$=>3\angle 0 = .33j \times (10\angle 45 + V2) - .167j \times V2 + .08 \times V2$

$=>3\angle 0 = 3.3\angle 135 + .33j \times V2 - .167j \times V2 + .08 \times V2$

$=>5.82\angle -23.63 = .33j \times V2 - .167j \times V2 + .08 \times V2$

$=>5.82\angle -23.63 = .163j \times V2 + .08 \times V2$

$=>5.82\angle -23.63 = V2 \times (.08 + .163j)$

$=>V2 = 32.1\angle -87.48$ **(Answer)**

So, $V1 = 10\angle 45 + 32.1\angle -87.48 = 26.351\angle -71.2377$ **(Answer)**

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